

WENDY TANG

Design Engineer | Robotics Researcher | Physical Hardware & HRI

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EDUCATION

Rhode Island School of Design (RISD) | BFA Industrial Design, 2026 — Robert O'Neal Creative Resourcefulness Award

FUTURE MOBILITY & ROBOTICS RESEARCH

HMG x RISD Regenerative Design Studio

— Future Mobility · Road Communication Platform Jun – Aug 2026

- Led user research on driver psychology, perception of abstract symbols, and communication breakdowns in traffic; analyzed data to set design direction
- Designed a road-communication platform expanding drivers' on-road vocabulary; surveys showed it outperformed existing signaling in empathy and road-rage reduction
- Applied AI tools to build a custom animation/UV-mapping interface; managed the project, delivering the final model and renderings on time for exhibition at Woods-Gerry Gallery

— Bipedal Robot Passive Leg Mechanism Feb 2026 – Present

- Designed and built a bipedal robot leg using passive multi-joint locking mechanisms derived from emu anatomy, reducing actuator count from 4–6 (industry standard) to 2
- Translated published biomechanics research (BirdBot, Max Planck Institute) into a simplified, fully documented physical prototype
- Achieved passive standing with zero motor engagement and a full walking gait cycle driven by two motors per leg
- Currently collecting performance data, testing under payload, and integrating IMU sensors for closed-loop control

— eVTOL Landing Gear · Team Research with Hyundai Motor Group Sep – Dec 2025

- Identified passive tendon-routing in avian anatomy as the key mechanism for terrain-adaptive grip; prototyped a landing system using a single actuator to control multi-DOF joints across varied surfaces
- Presented biological findings and eVTOL deployment scenarios to Hyundai engineering and design stakeholders

PROFESSIONAL EXPERIENCE

RISD x NASA Rover | Designer & Fabricator — 2nd Place Nationally Sep 2025 – Mar 2026

- Designed, fabricated, and tested the steering system for a human-powered rover at NASA HERC 2026
- Conducted a 4-criteria trade study and co-designed a four-bar linkage, improving ergonomics, durability, and fabrication simplicity
- Fabricated with carbon fiber layup and machined custom hardware; iterated steering geometry through real-terrain testing

HAVEN | Industrial Designer — AI + Elder Care Jan 2026 – Present

- Industrial designer on a cross-functional team of software and hardware engineers building an AI fall-detection camera for older adults
- Led user interviews and market research to define pain points and form language; drove ideation, visualization, and product narrative
- Designed a physical indicator that clearly shows users when the camera is on or off — balancing safety monitoring with privacy and dignity

Brown Tingle Lab | Lab Assistant — Locomotion Research Jan 2026 – Present

- Collected snake locomotion data via XMALab motion tracking; performed dissections comparing vertebral counts across snake species
- Trained in lab equipment handling; joined weekly research discussions and co-organized a science-to-art workshop with the Community College of Rhode Island

QTL | R&D Industrial Design Intern Jun – Aug 2025

- Engineered internal components for architectural lighting fixtures: universal hubs connecting multiple lens sizes while concealing wiring, louvers for targeted optical quality, and snap-fit LED covers for existing extrusion profiles; produced technical drawings for manufacturing

Wildlife Preservation | Studio Assistant – Physical Prototyping and Modeling Jun – Aug 2024

- Fabricated animal mannequins and diorama elements for natural history museums; conducted material testing for realism and durability

LEADERSHIP & COMMUNICATION

TEDx RISD Speaker (2025) — Presented on preservation, adaptation, and reconstruction of forms of life to a general audience.

RISD Taxidermy Club, Founder & President (2023–2025) — Built a cross-disciplinary community around natural history; established collaborations with the American Museum of Natural History, professional taxidermists, and biology Ph.D. researchers.

Teaching Assistant, RISD (2024) — Led discussions on human-technology relationships and AI; taught Rhino and digital fabrication.

SKILLS

Prototyping & Fabrication: Carbon fiber layup, metal machining (mill, lathe), additive manufacturing (FDM/SLA), silicone molding/casting, soldering, Arduino, rapid prototyping across mixed materials

CAD: SolidWorks, Fusion 360, Rhino + Grasshopper, Blender, KeyShot, FEA simulation

Bio-Inspired Design: Biomimicry framework, data analysis, motion data capture (XMALab), material property testing

Research & HRI: Human factors design, user research for robotics deployment, scientist collaboration, technical communication to non-specialists